UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	:	ICT450
EXAMINATION	:	NOVEMBER 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the ICT450 mock paper (November 2026). Answers are unofficial model answers, not official UTM marking schemes.

2. Marking points state the expected answer and the allocation of marks; every question retains its source paper and page reference.

3. Five questions x 20 marks; total = 100 marks.

4. Real-bank questions: 5. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 12 unique ICT450 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 7 printed pages.

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MARKING GUIDE

QUESTION 1 (20 marks)

Source: ICT450-2024-JAN · page [2] · bank id ICT450-2024-JAN-Q1

Reference answer

(a) Meaning of data and database. [4 marks]

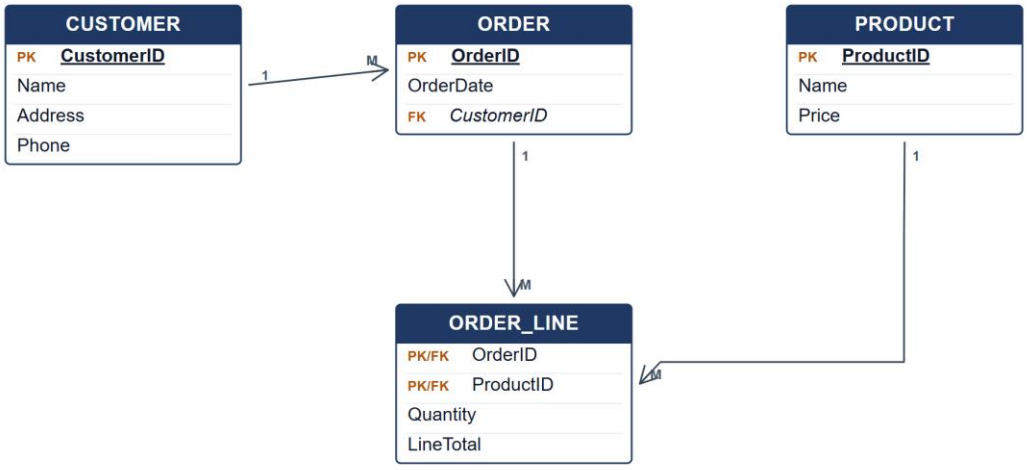
- Data are raw facts, such as a customer name or an order amount, that have not yet been processed into meaning.
- A database is an organised collection of logically related data stored and managed as a single repository.

(b) Data inconsistency and an example. [4 marks]

- Data inconsistency exists when the same fact has different values in different places.
- Example: a customer's phone number is updated in the Customers table but the old number remains in an Orders table.

(c) Zinnirah's problem and the ERD solution. [6 marks]

- The three tables are not linked by keys, so joins become complex and the same customer/product data is duplicated.
- The solution is an ERD with CUSTOMER, ORDER, PRODUCT and ORDER_LINE, where ORDER_LINE resolves the many-to-many order-product relationship.



ERD for the ZY Stationary database.

(d) Three types of SDLC maintenance. [6 marks]

- Corrective maintenance — fixing errors found after delivery.
- Adaptive maintenance — changing the system for a new environment, platform or business rule.
- Perfective maintenance — improving performance or adding enhancements requested by users.

Marking guide

- a) 2 marks each for data and database (4).
- b) 2 marks for the definition and 2 for the example (4).
- c) 2 marks for the problem and 4 marks for the ERD (6).
- d) 2 marks for each maintenance type (6).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the

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ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 1 DATABASE CONCEPT (1).pdf; TOPIC 2 DATA MODEL (1).pdf; TOPIC 3 RELATIONAL DATA MODELstu.pdf

Assumptions / notes

- The three tables are Customers, Orders and Products; an order can contain many products.

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QUESTION 2 (20 marks)

Source: ICT450-2023-JUL · page [2] · bank id ICT450-2023-JUL-Q2

Reference answer

(a) Four activities in the database design phase. [8 marks]

- Create the conceptual design — identify entities, attributes, relationships and business rules.
- Select the DBMS software — choose a platform that fits the data model, cost and hardware.
- Create the logical design — map the conceptual model to tables, keys, domains and constraints.
- Create the physical design — choose storage, indexes, file organisation and access paths.

(b) Mobile database suitability and three usage factors. [8 marks]

- Yes, a mobile database is suitable for a company with branches in Malaysia and Indonesia because staff need access away from the office.
- Factor 1: ubiquitous connectivity — field staff can reach the database from any branch or location.
- Factor 2: synchronization — local device data can be reconciled with the central database when connectivity returns.
- Factor 3: portability and location-aware services — mobile devices support on-site inventory and customer service.

(c) Structured versus unstructured data. [4 marks]

- Structured data has a predefined schema and is stored in rows and columns; example: a sales table.
- Unstructured data has no predefined relational schema; example: images, video, audio or free-text documents.

Marking guide

- a) 2 marks for each of four design activities (8).
- b) 2 marks for suitability and 2 marks for each of three factors (2 + 6 = 8).
- c) 2 marks for each data type with example (4).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 2 DATA MODEL (1).pdf; TOPIC 6 DATABASE DESIGN (1).pdf

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QUESTION 3 (20 marks)

Source: ICT450-2024-JUL · page [4] · bank id ICT450-2024-JUL-Q3

Reference answer

The Sasa eStore invoice contains a repeated item list and member data. It is flattened to one row per invoice line.

(a) Normal form of Figure 1. [3 marks]

- The invoice is in 1NF when flattened to one row per item; all values are atomic.
- It is not in 2NF because member details depend on MemberID, while item price depends on ItemName, not on the full invoice key.
- It is not in 3NF because MemberName and Address depend transitively on MemberID, and totals are repeated.

(b) Primary key and dependencies. [6 marks]

Flattened invoice relation.

```
INVOICE (InvoiceNo, InvoiceDate, MemberID, MemberName, Address, ItemName, Quantity, UnitPrice, Amount,
Total, PaidAmount, BalanceDue)
```

Functional dependencies.

```
InvoiceNo → InvoiceDate, MemberID, Total, PaidAmount, BalanceDue
MemberID → MemberName, Address
ItemName → UnitPrice
InvoiceNo, ItemName → Quantity, Amount
```

(c) Normalise from 1NF to 3NF. [11 marks]

1NF: one row per invoice item; key (InvoiceNo, ItemName).

2NF: remove partial dependencies.

2NF relations.

```
INVOICE (InvoiceNo, InvoiceDate, MemberID, MemberName, Address, Total, PaidAmount, BalanceDue)
ITEM (ItemName, UnitPrice)
INVOICE_ITEM (InvoiceNo, ItemName, Quantity, Amount)
```

3NF: remove MemberID → MemberName/Address.

3NF relations.

```
MEMBER (MemberID, MemberName, Address)
INVOICE (InvoiceNo, InvoiceDate, MemberID, Total, PaidAmount, BalanceDue)
ITEM (ItemName, UnitPrice)
INVOICE_ITEM (InvoiceNo, ItemName, Quantity, Amount)
```

Sample 3NF data: ITEM

ItemName	UnitPrice
Green Deep Lip & Eye Remover	65.00
Argan Oil Cleansing Foam	49.00
10% Niacinamide Serum	56.00

Marking guide

- a) 3 marks: 1NF plus 2NF/3NF reasons.
- b) 6 marks: key (2), dependencies (4).
- c) 11 marks: 1NF (2), 2NF (4), 3NF (5).

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Knowledge base basis: TOPIC 5 NORMALIZATION (1).pdf

Assumptions / notes

- The invoice has no printed item ID; ItemName is treated as the product identifier. The duplicate Argan Oil line is a repeated purchase line.

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QUESTION 4 (20 marks)

Source: ICT450-2025-FEB · page [4] · bank id ICT450-2025-FEB-Q4

Reference answer

e-Travel schema.

```
CUSTOMERS (CustomerID, FirstName, LastName, PhoneNumber)
BOOKING (BookingID, BookingDate, NumberOfPax, TripDate, CustomerID*, GuideID*, PackageID*)
GUIDE (GuideID, GuideName, BasicSalary, HireDate)
PACKAGE (PackageID, PackageName, PricePerPax, PackageDetails)
```

(a) Total number of trips for each date. [4 marks]

```
SELECT TripDate, COUNT(*) AS total_trips
FROM BOOKING
GROUP BY TripDate;
```

(b) Customer with the highest number of bookings. [4 marks]

```
SELECT c.*
FROM CUSTOMERS c
JOIN BOOKING b ON b.CustomerID = c.CustomerID
GROUP BY c.CustomerID, c.FirstName, c.LastName, c.PhoneNumber
ORDER BY COUNT(*) DESC
FETCH FIRST 1 ROW ONLY;
```

(c) Increase 10% salary for guides on a February 2025 trip. [4 marks]

```
UPDATE GUIDE
SET BasicSalary = BasicSalary * 1.10
WHERE GuideID IN (
    SELECT GuideID
    FROM BOOKING
    WHERE TripDate BETWEEN DATE '2025-02-01' AND DATE '2025-02-28'
);
```

(d) Packages booked more than 70 times. [4 marks]

```
SELECT p.PackageName
FROM PACKAGE p
JOIN BOOKING b ON b.PackageID = p.PackageID
GROUP BY p.PackageID, p.PackageName
HAVING COUNT(*) > 70;
```

(e) Customers who booked between February and March 2025. [4 marks]

```
SELECT COUNT(DISTINCT CustomerID) AS total_customers
FROM BOOKING
WHERE TripDate BETWEEN DATE '2025-02-01' AND DATE '2025-03-31';
```

Marking guide

- a) 4 marks: GROUP BY date and COUNT.
- b) 4 marks: join, GROUP BY customer, descending order/top-1.
- c) 4 marks: UPDATE, 10%, February subquery.
- d) 4 marks: join, GROUP BY package, HAVING > 70.
- e) 4 marks: COUNT DISTINCT and date range.

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 7 SQL.pdf

Assumptions / notes

- TripDate is the date used for the February/March filters.

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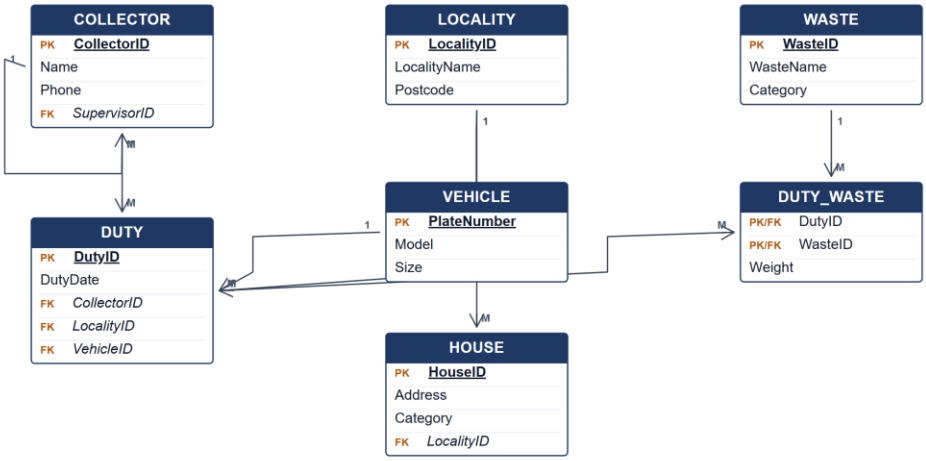
QUESTION 5 (20 marks)

Source: ICT450-2025-JUL · page [5, 6] · bank id ICT450-2025-JUL-Q5

Reference answer

The ERD models collectors, duties, localities, vehicles, houses and waste types. The supervisor relationship is recursive on COLLECTOR, and DUTY_WASTE records the weight collected for each waste type.

(a) Construct the ERD with Crow's Foot notation. [16 marks]



Clean Environment ERD.

(b) Two pieces of information from the ERD. [4 marks]

- The collection duties assigned to each collector, with date, locality and vehicle.
- The total weight collected for each waste type and locality.

Marking guide

- a) 16 marks: collector/duty/locality/vehicle (6); house and waste (4); recursive supervisor and DUTY_WASTE relationships (4); keys (2).
- b) 2 marks for each information item (4).

Provenance: model_unofficial · confidence medium · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 4 ERD_stu (2).pdf; TOPIC 3 RELATIONAL DATA MODELstu.pdf

Assumptions / notes

- Supervisor is a recursive role on COLLECTOR; a duty collects many waste types with a weight per waste type.

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